

The background is a dark, textured surface. A wavy, light-colored line, resembling a musical staff, curves across the frame. Several 3D musical notes are placed along this staff. From left to right, there is a sharp symbol (#), a single eighth note, a pair of beamed eighth notes, another single eighth note, a pair of beamed eighth notes, and a final single eighth note. The text "Music Store Data Analysis" is centered over the staff in a white, serif font.

Music Store Data Analysis

ABOUT ME

- **Background:** I recently completed my B.Tech in Computer Science Engineering (Data Science) from Sreyas Institute of Engineering and Technology. Throughout my studies, I developed a strong interest in Data Analytics and its applications in **data-driven decision-making**
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BUSINESS PROBLEM & ANALYTICAL OBJECTIVES

- **Executive Objective:** Optimize marketing budgets, boost retention, and maximize festival ROI using data-driven insights.
- **Operational Hierarchy:** Map top leadership to streamline internal escalation paths.
- **Market Expansion:** Identify top-grossing cities to select the best location for a promotional music festival.
- **VIP & Retention:** Pinpoint top-spending customers globally and regionally to launch targeted VIP programs.
- **Audience Segmentation:** Profile Rock genre listeners and artist affinity for cross-selling campaigns.
- **Catalog Optimization:** Analyze song durations to refine track bundling, playlists, and regional offerings.

AGENDA

1. Setup & Data Foundation

- Business context, objectives, and relational schema overview (11 tables, 3,500 tracks).

2. Operations & Revenue Distribution

- Management hierarchy, top revenue countries (USA, Canada, Brazil), and peak invoice values.

3. Core Customer & Catalog Drivers

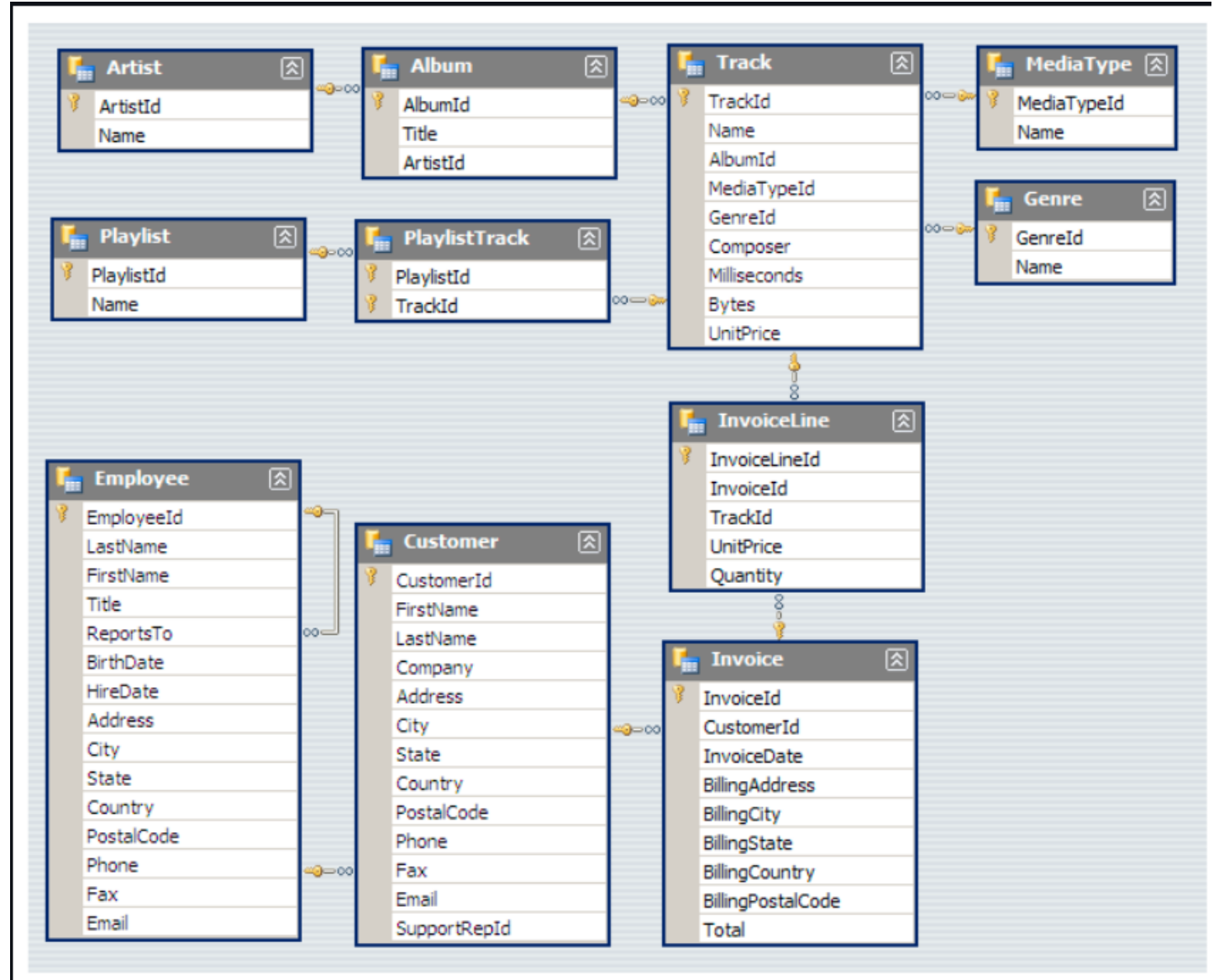
- Top festival market (Prague), leading VIP customer, and Rock catalog performance.

4. Advanced Segmentation & Strategic Roadmap

- Regional genre dominance, customer-artist cross-sell models, and actionable business next steps.

DATASET AND APPROACH

- **Database Scale:** Relational schema containing 11 normalized tables (Employees, Customers, Invoices, Tracks, Genres, Albums, Artists, Playlists).
- **Data Volume:** 3,500 tracks, 614 invoices, and 59 customers spanning 24 countries.
- **Analytical Stack:** MySQL utilizing multi-table joins, aggregations, Subqueries, Common Table Expressions (CTEs), and Window Functions (`ROW_NUMBER()` / `DENSE_RANK()`).



Team Structure & Operational Hierarchy

- **Leadership Role:** Madan Mohan serves as the Senior General Manager at level L6.
- **Hierarchy:** Holds the top organizational position with no direct supervisor.
- **Decision Authority:** Exercises ultimate sign-off on enterprise budget allocations and high-priority customer escalations.
- **Account Management:** Operational support representatives directly oversee customer accounts across all international territories.

GLOBAL REVENUE FOOTPRINT

- **Top 3 Markets by Order Volume:**

- **USA:** 131 invoices (market leader)
- **Canada:** 76 invoices
- **Brazil:** 61 invoices

	billing_country	Invoice_Count
▶	USA	131
	Canada	76
	Brazil	61

- **High-Value Single Transactions:** Top 3 individual invoice amounts are **\$23.76**, **\$19.80**, and **\$19.81**, driven primarily by large bundled track purchases.

	total	invoice_jd
▶	23.759999999999998	183
	19.8	92
	19.8	31
✱	NULL	NULL

THE FESTIVAL CITY

- **Top Revenue City:** Prague
- **Total Sales Volume:** \$273.24
- **Strategic Takeaway:** Prague delivers the highest urban revenue concentration.
- **Recommendation:** Allocate the primary event budget to Prague for the inaugural brand music festival to maximize local engagement and ROI.

BILLING_CITY	Invoice_Total
Prague	273.24000000000007

IDENTIFYING OUR HIGHEST-VALUE CUSTOMER

- **Top Customer:** František Wichterlová (Czech Republic).
- **Cumulative Lifetime Spend: \$144.54.**
- **Retention Strategy:** Establish a Tier-1 VIP loyalty program offering pre-sale access, merchandise discounts, and festival backstage passes.

	CUSTOMER_ID	FIRST_NAME	LAST_NAME	TOTAL_AMOUNT
▶	5	František	Wichterlová	144.54000000000002

THE ROCK AUDIENCE SEGMENT

- **Audience Reach:** 59 unique customers purchased Rock tracks, representing our most widespread demographic.
- **Top 10 Rock Bands by Catalog Size:**
 - **Led Zeppelin:** 114 tracks
 - **U2:** 112 tracks
 - **Deep Purple:** 92 tracks
 - **Iron Maiden:** 81 tracks
 - **Pearl Jam:** 54 tracks
 - **Van Halen:** 52 tracks
 - **Queen:** 45 tracks
 - **The Rolling Stones:** 41 tracks
 - **Creedence Clearwater Revival:** 40 tracks
 - **Kiss:** 35 tracks

	NAME	NUMBER_OF_SONGS
▶	Led Zeppelin	114
	U2	112
	Deep Purple	92
	Iron Maiden	81
	Pearl Jam	54
	Van Halen	52
	Queen	45
	The Rolling Stones	41
	Creedence Clearwater Revival	40
	Kiss	35

Marketing Angle: Create artist-specific email funnels and vinyl box-set cross-promotions for these core classic rock listeners.

CATALOG OPTIMIZATION & TRACK LENGTHS

- **Benchmark:** Average song duration across the entire catalog is ~**393.6 seconds (6.56 minutes)**.
- **Pattern Identified:** Outlier tracks (20 to 40+ minutes, primarily classical movements and live jam sessions) skew average catalog playtime.
- **Catalog Strategy:** Package extended-length recordings into dedicated deep-dive playlists and maintain standard pricing tiers for standard radio cuts.

	NAME	MILLISECONDS
▶	Occupation / Precipice	5286953
	Through a Looking Glass	5088838
	Greetings from Earth, Pt. 1	2960293

ARTIST SPEND & REGIONAL GENRE DOMINANCE

- **Top-Grossing Artist:** Queen drives the highest aggregate customer spend.
- **Cross-Sell Potential:** Pair Queen purchases with Led Zeppelin or Deep Purple at checkout to boost basket size.
- **Strategic Alignment:** Allocate 70% of digital ad spend to core Rock while localizing regional campaigns.

CUSTOMER_ID	FIRST_NAME	LAST_NAME	ARTIST_NAME	AMOUNT_SPENT
17	Jack	Smith	Queen	1.98
11	Alexandre	Rocha	Queen	1.98
46	Hugh	O'Reilly	Queen	27.719999999999985
1	LuÃ-s	GonÃ§alves	Queen	1.98
8	Daan	Peeters	Queen	1.98
52	Emma	Jones	Queen	1.98
23	John	Gordon	Queen	2.9699999999999998
5	FrantiÅiek	WichterlovÅi	Queen	3.96
48	Johannes	Van der Berg	Queen	1.98
16	Frank	Harris	Queen	1.98

TOP SPENDER PER COUNTRY

- **Key Global VIPs:**
 - **Czech Republic:** František Wichterlová (\$144.54)
 - **Ireland:** Hugh O'Reilly (\$114.839)
 - **India:** Manoj Pareek (\$111.869)
- **Account Strategy:** Assign dedicated customer support reps to top spenders in each territory to improve lifetime value and repeat purchase rates.

COUNTRY	CUSTOMER_NAME	TOTAL_SPENT
Czech Republic	František Wichterlová	144.54000000000002
Ireland	Hugh O'Reilly	114.83999999999997
India	Manoj Pareek	111.86999999999999
Brazil	Luís Gonçães	108.89999999999998
Portugal	João Fernandes	102.96000000000001
Canada	François Tremblay	99.99
France	Wyatt Girard	99.99
Spain	Enrique Muñoz	98.01

STRATEGIC RECOMMENDATIONS

- **Host Festival in Prague:** Direct event sponsorship budget to our highest revenue-generating city (\$273.24).
- **Launch Rock Retargeting Campaigns:** Re-engage all 59 Rock listeners with targeted classic rock bundles (Led Zeppelin, U2, Queen).
- **Deploy Global VIP Program:** Reward regional top spenders with bespoke loyalty perks and early catalog access.
- **Reorganize Streaming Playlists:** Separate standard radio-length tracks from 6.5+ minute long-form tracks to optimize recommendation algorithms.

THANK
YOU

